Coding Exercise 1

1. Generate $X_1, X_2, \ldots, X_{10000}$ i.i.d. $\sim P$, if
   (a) $P \equiv \text{binomial}(n, p)$,
   (b) $P \equiv \text{geometric}(p)$,
   (c) $P \equiv \text{negative binomial}(r, p)$.

   Obtain the parameters as input. Use only *numpy.random.rand* command.

2. Generate $X_1, X_2, \ldots, X_{10000}$ i.i.d. $\sim \text{Poisson}(\lambda)$ for a given $\lambda > 0$, using only *numpy.random.exponential* command.

3. For Q1 and Q2, verify if the generated data points follow the corresponding distribution using chi-squared test. Do this without using the *scipy.stats.chisquare command*. In your code, include the removal of data points that have $O_i > 0$ but $Ei \ll 1$.

4. Generate $X_1, X_2, \ldots, X_{10000}$ i.i.d. $\sim F$, if $F \equiv \text{Gamma}(3, \beta)$. Obtain β as input, and use only *numpy.random.exponential* command. Note that $\text{Gamma}(3, \beta)$ can be generated by adding three i.i.d. $\exp(\beta)$ random variables. Verify if the generated data points follow $\text{Gamma}(3, \beta)$ using Kolmogorov-Smirnov test. Do this without *scipy.stats.ks_1samp* or *scipy.stats.kstwo* commands.

5. (a) Fix the values of π, μ_1, σ_1^2, μ_2, σ_2^2 arbitrarily. Generate 10000 data points as follows. Pick $\Delta_i \sim \text{Bernoulli}(\pi)$, and pick $X_i \sim \mathcal{N}(\mu_1, \sigma_1^2)$ if $\Delta_i = 1$ and $X_i \sim \mathcal{N}(\mu_2, \sigma_2^2)$ if $\Delta_i = 0$ for all $i = 1, 2, \ldots, 10000$.
   (b) Assuming $X_1, X_2, \ldots, X_{10000}$ as the given data, estimate the parameters $\hat{\pi}$, $\hat{\mu}_1$, $\hat{\sigma}_1^2$, $\hat{\mu}_2$, $\hat{\sigma_2}^2$ using EM algorithm. Start by initializing $\pi = 0.5$, μ_1, μ_2 as random values, and σ_1^2, σ_2^2 as the sample variance.
   (c) Extend the code for a mixture of three normal distributions.

6. Repeat the above procedure when the distributions are $\text{Binomial}(20, p_1)$ and $\text{Binomial}(20, p_2)$. In M-step, fix $\hat{p}_i = \frac{1}{20} \sum_{i=1}^{10000} \left(\frac{\hat{\gamma}_i}{\sum_{i=1}^{10000} \hat{\gamma}_i} \right) x_i$.